

Question 1:

From previous chapters we know that number of params is :

$$\begin{aligned}\text{Params} &= (3DF + 2D(N+K) \cdot H) \cdot L + DV \\ &= (3 \cdot (4096) \cdot (4 \cdot 4096) + 2 \cdot 4096 \cdot (40) \cdot 256) \cdot 64 \\ &\quad + 4096 \cdot 32768 \\ &= 18385207246 \approx 18 \cdot 10^9 = 18 \text{ Billion}\end{aligned}$$

Per token KV cache size is :

$$\frac{2 \cdot S \cdot L \cdot K \cdot H}{S} = 2 \cdot L \cdot K \cdot H$$

here in int8 is :

$$\begin{aligned}2 \cdot 64 \cdot 8 \cdot 256 &= 262144 \text{ Bytes} \\ &\approx 262 \text{ KB.}\end{aligned}$$

Question 2:

Assume TPU use 4×4 slice where
use HBM = 16 GB.

Note that:

$$\text{Space per shard} = \frac{M}{|X|} + \frac{B \cdot S \cdot L \cdot K \cdot H}{|X|}$$

$$\frac{M}{|X|} + \frac{B \cdot S \cdot L \cdot K \cdot H}{|X|} \leq \text{HBM}$$

where M is Total Params in int8.

Therefore the largest batch size we can fit in is:

$$\begin{aligned} \frac{M}{|X|} + \frac{B \cdot S \cdot L \cdot K \cdot H}{|X|} &\leq \text{HBM} \Leftrightarrow \frac{M}{|X|} - \text{HBM} \leq -\frac{B \cdot S \cdot L \cdot K \cdot H}{|X|} \\ \Leftrightarrow \frac{|X| \cdot \text{HBM} - M}{|X|} &\geq \frac{B \cdot S \cdot L \cdot K \cdot H}{|X|} \Leftrightarrow \frac{|X| \cdot \text{HBM} - M}{S \cdot L \cdot K \cdot H} \geq B \end{aligned}$$

For $S = 128K$ we get

$$B \leq \frac{16 \cdot (16 \cdot 10^4) - 18 \cdot 10^4}{2 \cdot 128 \cdot 10^3 \cdot 64 \cdot 8 \cdot 256}$$
$$= 7.09 \approx 7.$$

And if we let $K = 1$ we get:

$$B \leq \frac{16 \cdot 16e4 - 18e4}{2 \cdot 128e3 \cdot 64 \cdot 1 \cdot 256}$$
$$\approx 56.$$

Question 3:

Assume TPU use 4×4 slice where

VSe BW = $8.2e11$. Therefore to load

$\frac{M}{|X|}$ into the MXU it takes:

$$\frac{M}{|X|} \cdot \frac{1}{BW} = \frac{18e4}{16} \cdot \frac{1}{8.2e11} \approx 0.0013 \text{ s}$$
$$= 1.3 \text{ ms}.$$

Question 4:

Assume TPU use 4×4 slice.

With this slice we don't have wraparound links. To shard it both for prefill and generation, since during generation FSDP does not help, we could use tensor parallelism sharded over all the devices for both prefill and generation.

For prefill we are compute bound when:

$$\gamma < F \cdot \frac{W_{IC1}}{C} \cdot 2$$

$$\text{Let } F = 4.4096, \quad W_{IC1} = 9e10$$

and $C = 3.94e14$. Hence:

$$\gamma < 16384 \cdot \frac{9e10}{3.94e14} \cdot 2 \approx 7.5$$

so we cannot do tensor parallelism with 16 devices and be compute bound.

We could insted do TP an CP with 4 way TP and 4 way CP, to be compete bound, So then a roofline bound will be :

$$\epsilon = \frac{4 \cdot B \cdot D \cdot F}{X \cdot Y \cdot C} \quad \text{where } B = 7 \cdot 128K$$

During generation we will be memory bound insted of communication bound when using tensor parallelism when;

$$Y < F \cdot \frac{1}{B} \cdot \frac{W_{IC1}}{W_{HBM}}$$

where $\frac{W_{IC1}}{W_{HBM}} \approx \frac{1}{8}$ in a TPV use.

Hence:

$$Y < 4 \cdot 4096 \cdot \frac{1}{B} \cdot \frac{1}{8} = \frac{2048}{B}$$

Since for $S = 128K$ the maximum Batch size is 7 then we get

$$Y < \frac{2048}{7} = 292$$

therefore during generation we will be memory bound and a roofline bound will be:

$$t = \frac{2DF}{X \cdot W_{HBM}}$$

We can shard the KV caches as:

$$KV[2, B_x, S, K_y, H]$$

where the mesh is defined as $\{x:2, y:8\}$

A rough per step latency under this sharding is:

$$\begin{aligned} & \frac{B \times KV \text{ Cache Size} + \text{Parameter size}}{16 \cdot W_{HBM}} \\ &= \frac{7 \times (2 \cdot 128e3 \cdot 64 \cdot 8 \cdot 256) + 18e9}{16 \cdot 8.2e11} \approx 0.02s \end{aligned}$$

Question 5:

5.1

Let $E=16$ and $K=2$ then:

$$\begin{aligned}\text{Total Params} &= (3 EDF + 2 D(N+K) \cdot H) \cdot L + DV \\ &\simeq 212 \text{ Billion.}\end{aligned}$$

$$\begin{aligned}\text{Activated Params} &= (3 KDF + 2 D(N+K) \cdot H) \cdot L + DV \\ &\simeq 31 \text{ Billion}\end{aligned}$$

5.2

From the approximation of the MLP part they use in this chapter, to be flops bound we need $T_{\text{math}} > T_{\text{comms}}$:

$$\frac{2 \cdot B \cdot 3KDF}{C} > \frac{2 \cdot 3 \cdot EDF}{BW} \Leftrightarrow \frac{2 \cdot B \cdot \cancel{3KDF}}{\cancel{2 \cdot 3 \cdot EDF}} > \frac{C}{BW}$$

$$\Leftrightarrow B \cdot \frac{K}{E} > \frac{C}{BW} \Leftrightarrow B > \frac{E}{K} \cdot \frac{C}{BW}$$

5.3

KV cache size per token does not change

5.4

Using the formula from chapter 4:

$$\text{FLOPs} = 6 \cdot BT (3KDF + 2 \cdot D(N+K) \cdot H) \cdot L$$

Question 6:

6.1

Assuming we shard the MLPs as:

$$W_{in1} [E_z, D, F_y], W_{in2} [E_z, D, F_y]$$

$$W_{out} [E_z, F_y, D]$$

the attention params as:

$$W_q [D, N_y, H], W_k [D, K_y, H], W_v [D, K_y, H]$$

$$W_o [N_y, H, D]$$

and the shared input-output embedding as:

$$V [V_y, D]$$

The total params per device is:

$$\text{Params Per Device} = \left(3 \frac{E}{|Z|} \frac{DF}{|Y|} + 2 \frac{D}{|Y|} (N+K)H \right) \cdot L + D \frac{V}{|Y|}$$

$$\simeq 2.3 \text{ Billion}$$

Hence the weight loading time in bf16:

$$\frac{2 \cdot 2.3 \times 10^9}{8.2 \times 10^{11}} \simeq 0.005 \text{ s} = 5 \text{ ms}$$

As free space we have $\simeq 16 \text{ GB} - 4.6 \text{ GB} = 11.4 \text{ GB}$.

6.2

Fit $Z = 16$ so that we can shard our
expert copies, then we need to choose Y so that:

$$2 \cdot \frac{1}{|Y|} \left(3 DF + 2 D (N+K) \cdot H \right) \cdot L + DV \leq 16$$

$$\Leftrightarrow \frac{2 \cdot 18 \times 10^9}{16} \leq |Y| \Leftrightarrow |Y| \geq 2.25$$

hence if we need a slice with axis sizes being multiple of 2, the smallest slice we could fit our model on is 4×16 .

Question 7:

Let $N = |X| \cdot |Y| \cdot |Z|$ be the total number of devices in the slice. From the algorithm we can see that:

$$\begin{aligned} T_{\text{comms}} &= \frac{2BD}{|X|} \cdot \frac{1}{2 \cdot W_{ICI}} + \frac{2 \cdot 2 \cdot B \cdot F}{|Y| \cdot |Z|} \cdot \frac{1}{W_{ICI}} \\ &+ \frac{2BD}{|X|} \cdot \frac{1}{2 \cdot W_{ICI}} = \frac{2BD}{|X| \cdot W_{ICI}} + \frac{4BF \cdot |X|}{N \cdot W_{ICI}} \\ &= \frac{1}{W_{ICI}} \cdot \left(\frac{2BD}{|X|} + \frac{4BF|X|}{N} \right) \end{aligned}$$

$$\begin{aligned} T_{\text{math}} &= 2 \cdot B \cdot \frac{D}{|X|} \cdot \frac{F}{|Y| \cdot |Z|} \cdot \frac{1}{C} + 2 \cdot B \cdot \frac{F}{|Y| \cdot |Z|} \cdot \frac{D}{|X|} \cdot \frac{1}{C} \\ &= \frac{4BD F}{|X| \cdot |Y| \cdot |Z| \cdot C} = \frac{4BD F}{N \cdot C} \end{aligned}$$

Note that given N devices T_{comms} can change based in how we allocate devices for the different axis x, y, z . Therefore we need to find the setting where T_{comms} is minimum.

$$T_{\text{comms}}(|x|) = \frac{1}{W_{ICI}} \cdot \left(\frac{2BD}{|x|} + \frac{4BF|x|}{N} \right)$$

$$\frac{\partial T_{\text{comms}}}{\partial |x|} = \frac{1}{W_{ICI}} \cdot \left(2BD \cdot (-1) \cdot \frac{1}{|x|^2} + \frac{4BF}{N} \right)$$

Solving $\frac{\partial T_{\text{comms}}}{\partial |x|} = 0$ we get a candidate for a stationary point:

$$\frac{4BF}{N} = \frac{2BD}{|x|^2} \Leftrightarrow \frac{N}{4BF} \cdot 2BD = |x|^2$$

$$\Leftrightarrow |x|^2 = \frac{D}{2F} \cdot N$$

since $|x|$ can only be positive in our case

we get $x^* = \sqrt{\frac{D}{2F} \cdot N}$. let $F = 4D$ to
get $x^* = \sqrt{\frac{N}{8}}$.

Since $\frac{\partial^2 T_{\text{comms}}}{\partial |x|^2} \Big|_{|x|=x^*} > 0$ we know

that x^* is a global minimum and

T_{comms} becomes:

$$T_{\text{comms}} = \frac{1}{W_{ICI}} \left(\frac{2BD\sqrt{8}}{\sqrt{N}} + \frac{4BF\sqrt{N}}{N\sqrt{8}} \right)$$

$$= \frac{1}{W_{ICI}} \left(\frac{2BD\sqrt{8}}{\sqrt{N}} + \frac{4BF}{\sqrt{N} \cdot \sqrt{8}} \right)$$

$$= \frac{1}{W_{ICI}} \left(\frac{16BD}{\sqrt{8} \cdot \sqrt{N}} + \frac{4BF}{\sqrt{N} \cdot \sqrt{8}} \right)$$

$$= \frac{1}{\sqrt{8} \cdot W_{ICI}} \left(\frac{4B \cdot (4D + F)}{\sqrt{N}} \right)$$

$$= \frac{1}{\sqrt{8} \cdot W_{ICI}} \left(\frac{4B8D}{\sqrt{N}} \right) = \frac{1}{\sqrt{8} \cdot W_{ICI}} \cdot \frac{32BD}{\sqrt{N}}$$

$$\frac{T_{\text{math}}}{T_{\text{comm}}} > 1 \Leftrightarrow \frac{4 B D F}{N \cdot C} \cdot \frac{\sqrt{8} W_{IC1} \cdot \sqrt{N}}{32 B D} > 1$$

$$\Leftrightarrow \frac{4 \sqrt{8} \cdot W_{IC1} \cdot \sqrt{N} \cdot F}{32 N \cdot C} > 1$$

$$\Leftrightarrow \frac{4 \sqrt{8}}{32} \cdot \frac{1}{\sqrt{N}} \cdot \frac{W_{IC1} \cdot F}{C} > 1$$

$$\Leftrightarrow \frac{16 \cdot 8}{32^2} \cdot \frac{1}{N} \cdot \frac{W_{IC1}^2 \cdot F^2}{C^2} > 1$$

$$\Leftrightarrow \frac{16 \cdot 8}{32^2} \cdot \frac{F^2}{\alpha^2} > N$$

$$\Leftrightarrow \frac{1}{8} \left(\frac{F}{\alpha} \right)^2 > N$$

If we compare to traditional 3D model sharding where the condition to be compute bound is :

$$M_y \cdot \frac{F}{\alpha} > N$$

we see that with 2D weight stationary:

$$\frac{\frac{1}{8} \left(\frac{F}{\alpha}\right)^2}{M_{xyz} \left(\frac{F}{\alpha}\right)} = \frac{1}{8M_{xyz}} \cdot \left(\frac{F}{\alpha}\right)$$

During inference however if we are communication bound in the MLP because of a small batch size, we see that:

$$\begin{aligned} \frac{T_{\text{comms-2D}}}{T_{\text{comms-3D}}} &= \frac{32}{\sqrt{8}} \frac{BD}{W_{ICI} \cdot \sqrt{N}} \cdot \frac{W_{ICI} \cdot M_{xyz}}{4BD} \\ &= \frac{32}{\sqrt{8} \cdot 4} \cdot \frac{M_{xyz}}{\sqrt{N}} = 2 \cdot \sqrt{2} \cdot \frac{M_{xyz}}{\sqrt{N}} \end{aligned}$$